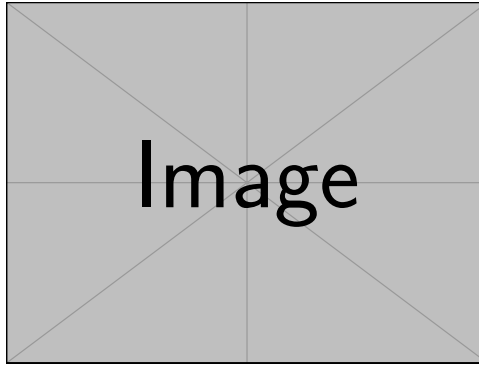




OCR A-Level Computer Science

H446/01: Computer Systems

Section 1.3: Exchanging Data

Practice Examination Paper

Time allowed: 75 minutes	
Total marks: 40	

Centre Number	
Candidate Number	
Surname	
Other Names	
Date	

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above.
- Answer ALL questions.
- Write your answers in the spaces provided.
- Additional paper may be used if required but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 40.
- Quality of extended responses will be assessed in questions marked with an asterisk (*).

Questions

1. A school uses a local area network to connect computers in its IT rooms.

(a) Define what is meant by a local area network (LAN). [2 marks]

(b) Describe two benefits of using a star topology in a school network compared to a bus topology. [4 marks]

(c) The school is considering implementing a wireless network.
Explain two security issues that could arise from implementing a wireless network, and how each issue could be addressed. [6 marks]

2. A company uses a client-server network model for its business operations.

(a) Explain what is meant by a client-server network model. [2 marks]

(b) Describe two advantages of a client-server network model compared to a peer-to-peer network model. [4 marks]

(c) The company's server uses TCP/IP protocols for network communication.
Explain the purpose of each of the following protocols:

(i) SMTP [2 marks]

(ii) FTP [2 marks]

(iii) HTTP [2 marks]

3. A user is sending an email with a large attachment to a colleague.

(a) Describe the process of packet switching when sending data across a network. [3 marks]

(b) Explain why data might need to be compressed before being sent across a network. [2 marks]

(c) The email is sent using the SMTP protocol.
Explain how the email reaches its destination, including the role of DNS servers. [4 marks]

4. A company is implementing encryption for its network communications.
- (a) Define the term 'encryption' and explain why it is important for network communications. [3 marks]

 - (b) The company is considering using symmetric key encryption.
 - (i) Describe how symmetric key encryption works. [2 marks]

 - (ii) Explain one disadvantage of symmetric key encryption compared to asymmetric key encryption. [2 marks]

 - (c) Explain how a MAC address is used in network communications. [2 marks]

5. (a) Explain the purpose of the following components in a network:
- (i) Router [2 marks]
- (ii) Switch [2 marks]
- (iii) Wireless Access Point [2 marks]
- (b) A company is considering implementing a web server for its internal intranet. Discuss the factors that the company should consider when implementing this server. [6 marks]

Grade Boundaries

Grade	A*	A	B	C	D
Marks	36-40	32-35	28-31	24-27	20-23